Inertia and balanced forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict motion when the net force is zero.

SUCCESS CRITERIA - I can...

1. Add forces to decide whether they balance.
2. Predict constant velocity when net force is zero.
3. Explain why changing direction requires net force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A goes 6 m east and 4 m north. B then goes 3 m west and 8 m south. Draw the route and find the endpoint relative to the start.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: ΣF = 0 ⇒ a = 0 ⇒ v = constant.
ΣF means net force. Balanced forces allow rest or steady motion in a straight line.

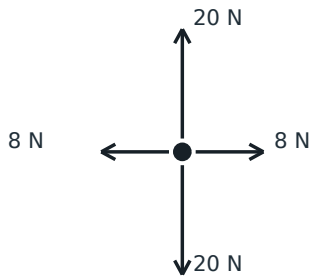
Worked reasoning: A book has 6 N upward support and 6 N downward weight. Net force is zero. Its velocity stays constant; it could be resting or moving steadily.

Your turn: Draw the two force arrows. Then sketch two possible velocity-time graphs consistent with this balance.

Marzano check - circle: ① ② ③

03 Balanced while moving

The box is sliding right at steady velocity. Add the horizontal forces shown. Does zero net force mean the box must stop?



The dot represents the object. Box moving right

Marzano check - circle: ① ② ③

04 After release

A hand pushes a puck, then loses contact. On an ideal frictionless level surface, what happens to the hand force and the puck's horizontal motion?

Marzano check - circle: ① ② ③

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Continue down each column.

05 The bus stops

Explain why a passenger tends to move forward when a bus brakes. Name what can supply the force that slows the passenger.

Marzano check - circle: 1 2 3

06 Choose a method

A cart accelerates uniformly from rest to 10 m/s east in 3 s. Find acceleration, average velocity and displacement.

Marzano check - circle: 1 2 3

07 Exit check

Which motions can have zero net force: resting; straight at steady speed; turning at steady speed; speeding up? Explain one choice.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A puck slides right at steady speed. Draw one possible balanced force diagram and explain why a forward force is not required in the frictionless model.

Marzano check - circle: 1 2 3